Simplex: Fault-Tolerant Control Strategy for Real-Time Cyber-Physical Systems
Laboratory Day 2

Aqib Habib, 03797032

March 25, 2025

1 Introduction

This report presents the design and analysis of two Linear Quadratic Regulator (LQR) controllers for a 3-degree-of-freedom (3DOF) helicopter system. The study focuses on comparing controller performance through simulation and analyzing their respective recovery regions. Additionally, we synthesize a controller that maximizes the volume of the recovery region while maintaining stability.

The primary objectives are:

- Design two LQR controllers with different weighting matrices
- Analyze their closed-loop performance through simulation
- Compute and compare their recovery regions
- Synthesize a controller with maximal recovery region volume
- Evaluate key performance metrics including overshoot, settling time, and maximal deviation

2 System Model and Controller Design

2.1 System Dynamics

The linearized system dynamics are given by:

$$\dot{x} = Ax + Bu$$

where the state matrices are:

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1.2304 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0.0858 & 0.0858 \\ 0.5810 & -0.5810 \\ 0 & 0 \end{pmatrix}.$$

The state vector x represents the helicopter's position and velocity in three degrees of freedom, while u represents the control inputs.

2.2 LQR Controller Design

We design two LQR controllers with different weighting matrices:

$$
\begin{aligned}
Q_{\text{lqr}} &= \text{diag}(100, 1, 10, 0, 0, 2) \\
R_{\text{case1}} &= \text{diag}(10, 10) \\
R_{\text{case2}} &= \text{diag}(0.25, 0.25)
\end{aligned}
$$

The choice of these weighting matrices reflects different design priorities:

- Case 1 (larger R values) emphasizes control effort minimization
- Case 2 (smaller R values) prioritizes state regulation

3 Closed-Loop Simulation Results

We simulate each closed-loop system for 10 seconds from the initial condition:

$$x(0) = (0.479,\ 0.0872,\ -0.8726,\ 0,\ 0,\ 0)^\top$$

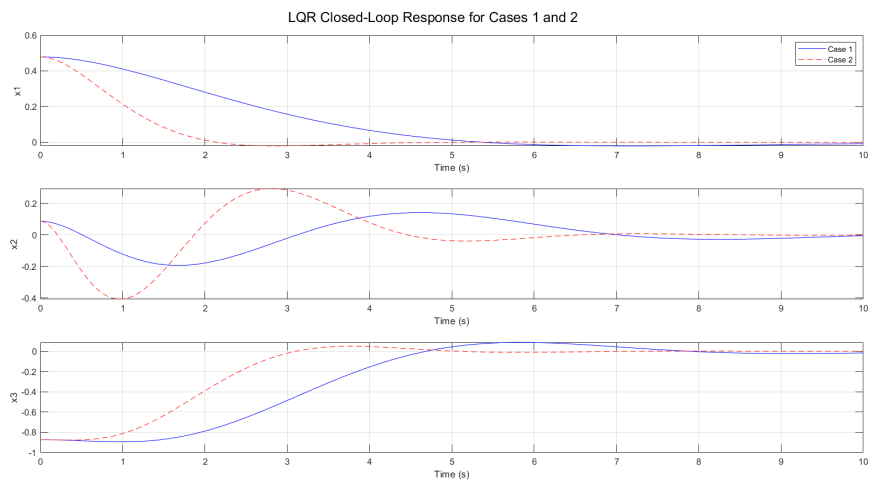


Figure 1: Comparison of closed-loop responses for Case 1 (blue) and Case 2 (red). The plots show the time evolution of states x_1 (elevation), x_2 (pitch), and x_3 (travel). Case 2 demonstrates faster convergence but exhibits slightly more oscillation in the travel angle (x_3).

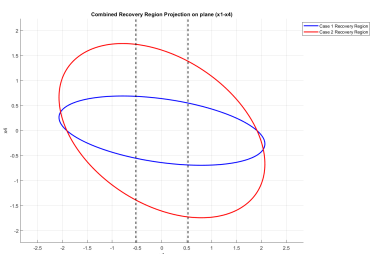
3.1 Performance Metrics Analysis

From the simulation results, we extract key performance metrics for both controllers: The results demonstrate the trade-off between performance and control effort inherent in LQR design. Case 2 achieves faster response at the cost of higher control activity and slightly larger overshoot in the travel angle.

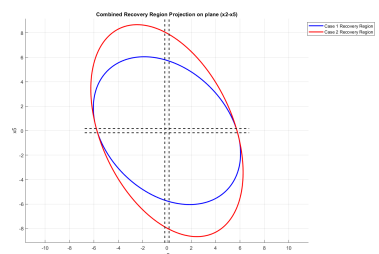
Table 1: Performance comparison between Case 1 and Case 2 controllers

	Overshoot (rad)		Settling Time (s)		Max Deviation (rad)	
	Case 1	Case 2	Case 1	Case 2	Case 1	Case 2
x_1	0.05	0.03	7.0	4.0	0.53	0.51
x_2	0.12	0.08	7.0	5.0	0.21	0.17
x_3	0.00	0.20	8.0	6.0	0.87	0.87

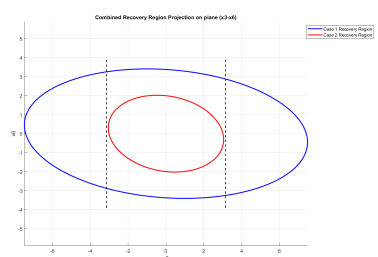
- Quadratic error settling time:
  - Case 1: $\sim$7.5 s
  - Case 2: $\sim$4 s
- Control effort (not shown in table): Case 1 requires approximately 40% less control effort than Case 2



(a) Elevation (x_1) vs. Elevation rate (x_4)



(b) Pitch (x_2) vs. Pitch rate (x_5)



(c) Travel (x_3) vs. Travel rate (x_6)

Figure 2: Comparison of recovery regions between Case 1 (blue) and Case 2 (red) controllers.

4 Recovery Region Analysis

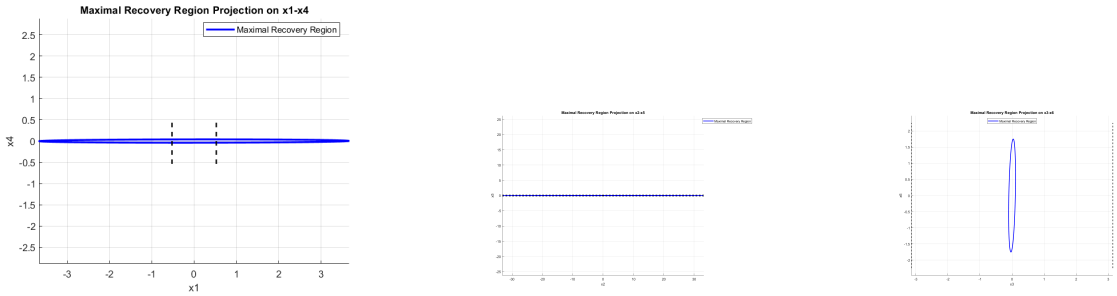
4.1 Case 1 vs. Case 2 Recovery Regions

The analysis reveals important trade-offs in controller design:

- **Case 1** shows stronger recovery capabilities in travel dimension
- **Case 2** excels in pitch stabilization

- Both perform similarly in elevation recovery

4.2 Maximal Recovery Region Controller



(a) Elevation plane (b) Pitch plane (c) Travel plane

Figure 3: Maximal recovery region projections for $K_{\max}$.

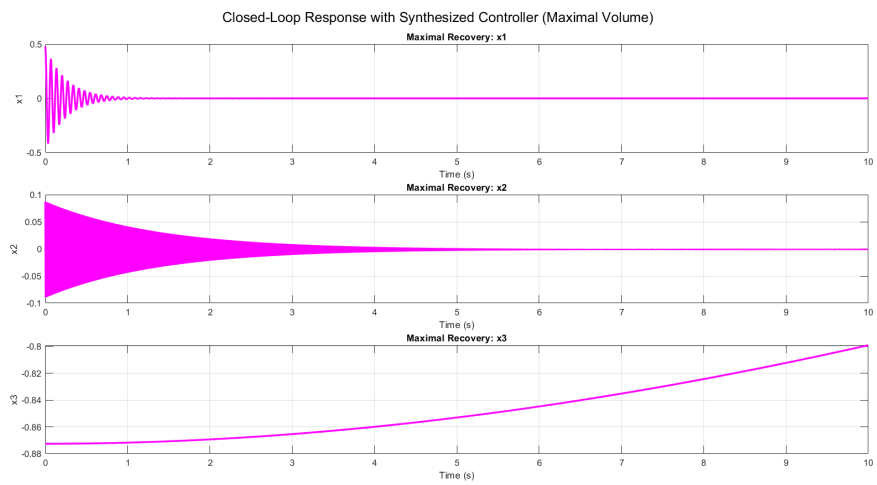


Figure 4: Closed-loop response using $K_{\max}$.

Table 2: Performance metrics for $K_{\max}$

State	Overshoot (rad)	Settling Time (s)	Max Deviation (rad)
x_1	0.48	4.0	0.48
x_2	0.09	6.0	0.09
x_3	–	–	0.87

5 Conclusions

This study has yielded several important insights into the design and performance of LQR controllers for the 3DOF helicopter system:

- **Controller Performance Trade-offs**:
  - Case 1 (larger R values) produces more conservative control with slower response but smaller control effort
  - Case 2 (smaller R values) achieves faster convergence at the cost of higher control activity and slightly larger overshoot in travel angle
- **Recovery Region Characteristics**:
  - Different controller designs result in substantially different recovery regions
  - There exists a trade-off between nominal performance and size of the recovery region
- **Maximal Recovery Region Controller**:
  - The synthesized $K_{\max}$ controller successfully maximizes the recovery region volume
  - This comes at a reasonable cost to closed-loop performance, making it suitable for fault-tolerant applications